

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the OEM service manual.

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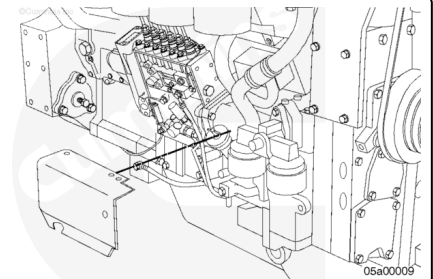


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⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

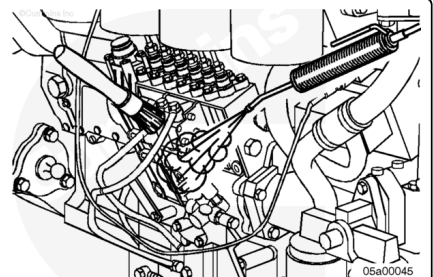
- Clean the debris from around the in-line fuel injection pump.
- Remove the open-drive coupling cover.



Remove

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

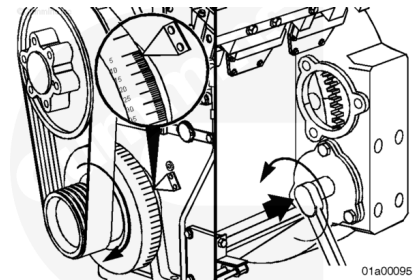
⚠ CAUTION ⚠

Use a cleaning solvent that will not harm aluminum.

Use a brush and solvent to clean the fuel injection pump.

⚠ CAUTION ⚠

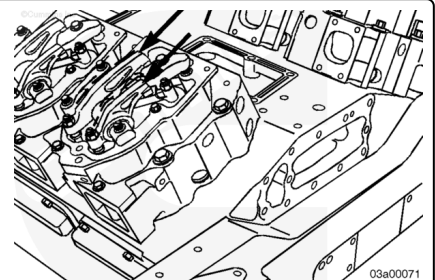
Both the right and left-bank fuel pumps must be removed/installed with the engine in this position. Severe engine damage can occur if static injection timing is not set properly.



Rotate the engine slowly clockwise until the crankshaft damper pointer aligns with the specified engine timing, which is provided on the engine dataplate, before Top Dead Center for the right - bank number 1 cylinder on the compression stroke.

Remove the right - bank number 1 rocker cover. Verify the number 1 cylinder is on the compression stroke and at specified timing before Top Dead Center.

Note : Both the rocker levers will be loose, or the push rods will be the same height, if the piston is on the compression stroke.

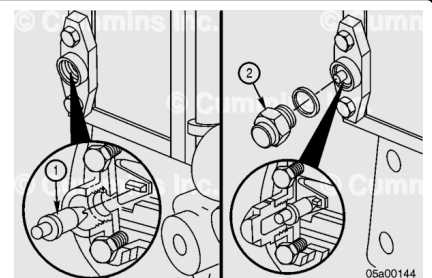


Remove the fuel injection pump timing pin cap (2). Verify the position of the timing pin. The fuel pump must be locked. Reverse the position of the timing pin (1) to lock the pump, as shown.

Install the timing pin, cap, and sealing washer.

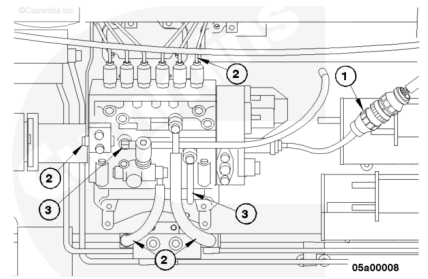
Torque Value: 30 N·m [22 ft-lb]

Note : If the pump is to be reused, the timing pin must set in the locked position to preserve the pump timing.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Disconnect the electrical connections. (1)

Disconnect the low and high-pressure fuel lines (2). Refer to Procedure 006-024 in Section 6.

(/qs3/pubsys2/xml/en/procedures/57/57-006-024-tr.html)

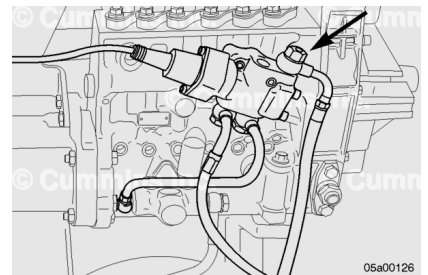
Remove the oil line. (3)

Install a protective cover to prevent any dirt or dust from getting into the discharge port of the injection pump or inlet port of the nozzle connector.

Remove the overflow/pressure regulator valve from the mechanical fuel pump. Identification, removal, inspection, clean for reuse, and assembly of the overflow/pressure valve is found in the following procedure.

Refer to Procedure 005-066 in Section 5.

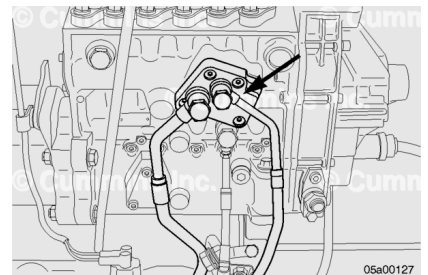
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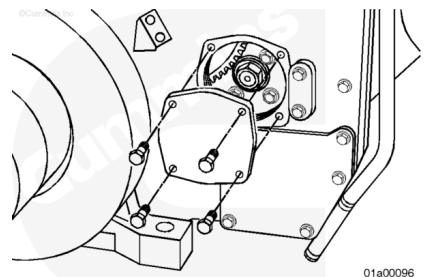
Remove the overflow/pressure regulator valve from the electronic fuel pump. Identification, removal, inspection, clean for reuse, and assembly of the overflow/pressure valve is found in the following procedure.

Refer to Procedure 005-066 in Section 5.

(/qs3/pubsys2/xml/en/procedures/57/57-005-066.html)

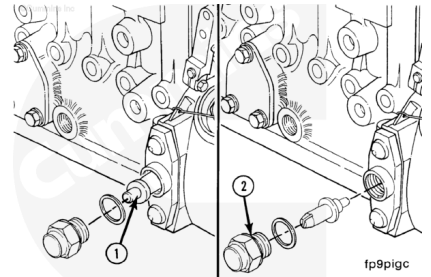


If equipped, remove the four capscrews securing the fuel pump drive gear cover plate. Remove the fuel pump drive gear cover plate and the cover plate o-ring.



Remove the fuel injection pump timing pin cap (2). Reverse the position of the timing pin (1) to unlock the pump and install the pin, cap and sealing washer.

Torque Value: 30 n•m [22 ft-lb]

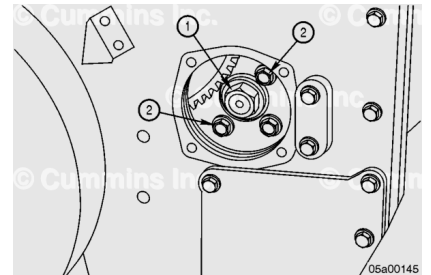


⚠ CAUTION ⚠

Severe damage to the fuel pump may occur unless the fuel pump is unlocked prior to breaking free the drive gear assembly from the drive shaft.

Break loose the large nut (1) and washer securing the drive gear assembly to the drive shaft.

Note : For a two piece gear assembly: Remove the two capscrews (2) securing the drive gear to the hub, as shown.

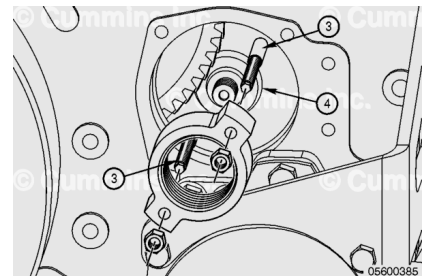


One-piece gear assembly

Thread and Hand tighten the remover screws (3) from the short threaded side on the drive gear and hub assembly (4).

Slide the mounting plate (1) on the remover screws (3). Until it contacts the flat on the remover screws.

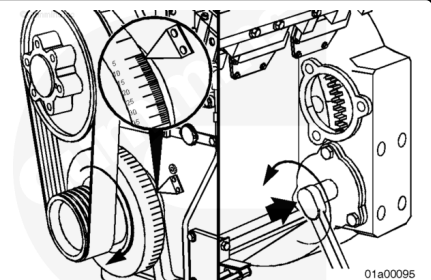
Thread and hand tighten the nuts (2) to make the mounting plate (1) seat against the flat on the remove screws.



⚠ CAUTION ⚠

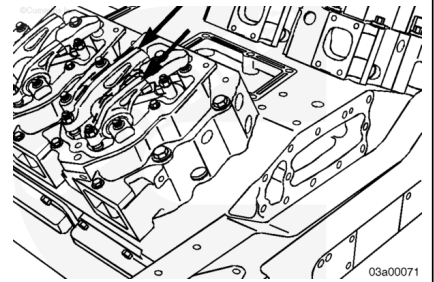
Both the right and left-bank fuel pumps must be removed/installed with the engine in this position. Severe engine damage can occur if static injection timing is not set properly.

Rotate the engine slowly clockwise until the crankshaft damper pointer aligns with the specified engine timing, which is provided on the engine dataplate, before top dead center for the right-bank number 1 cylinder on the compression stroke.



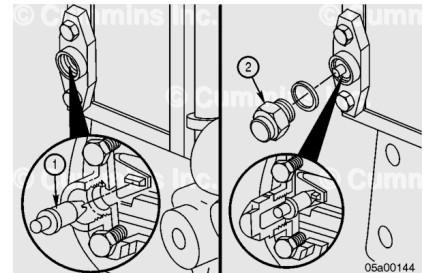
Remove the right-bank number 1 cylinder rocker cover. Verify the number 1 cylinder is on the compression stroke at specified timing before Top Dead Center.

Note : Both rocker levers will be loose or the push rods will be the same height if the piston is on the compression stroke.



Verify the engine is in the correct position for fuel pumps to be removed from the engine.

Remove the fuel injection pump timing pin cap (2). Verify the position of the timing pin. The fuel pump must be locked. Reverse the position of the timing pin (1) to lock the pump, as shown.



Install the timing pin, cap, and sealing washer.

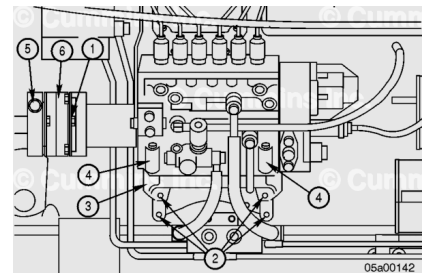
Torque Value: 30 n·m [22 ft·lb]

Note : If the pump is to be reused, the timing pin must set in the locked position to preserve the pump timing.

Remove the injection pump assembly:

Open-Drive (Base-Mount) Fuel Pump (P8500)

- Disconnect the coupling bolts (1); **only** loosen the larger bolt (5).
- Fabricate a spreader bolt and use it to loosen the pinch plate (6).
- Loosen the four injection pump bracket mounting bolts (2).
- Remove the injection pump and bracket (3) at an angle as one unit.
- Remove the mounting bolts (4).

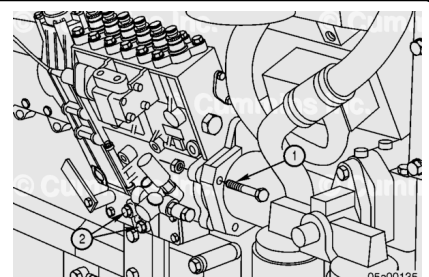


Note : The removal of the drive coupling from the injection pump is the responsibility of Bosch. Contact a Bosch™ dealer for instructions before delivering the pump to the repair facility.

Flange-Mount Fuel Pump (P8500, P7100, and RP39)

- Remove the four capscrews(1), washers, and nuts holding the pump to the drive housing.
- Remove the injection pump bracket mounting bolts (2).

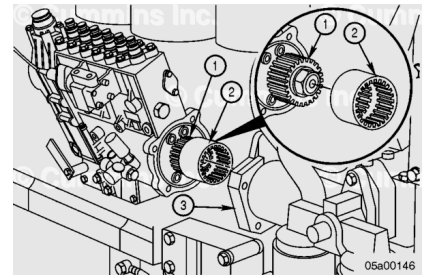
Note : The right-bank fuel pump bracket has six mounting bolts. The left-bank pump bracket has three



mounting bolts.

To remove the pump, slide the pump to the rear and disengage the spline coupling (2) from the drive housing (3) and the pump driveshaft spline adapter (1) from the spline coupling (2).

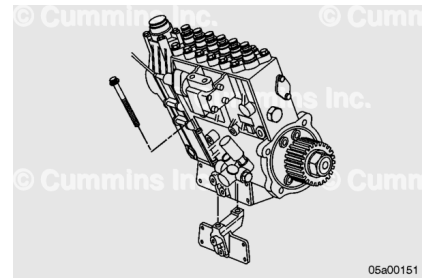
Remove the pump from the drive housing.



Remove the mounting bolts and the fuel pump bracket.

The pump driveshaft spline adapter **MUST** be removed and kept if the existing pump needs to be replaced or repaired. Replacement fuel injection pumps do **NOT** include spline adapters.

See instructions below for removal of the driveshaft spline adapter.

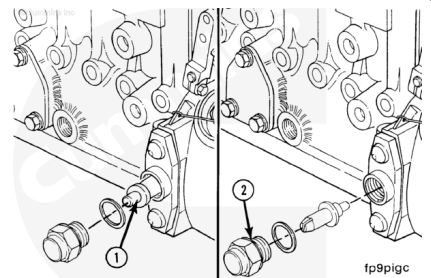


⚠ CAUTION ⚠

Check to make sure the injection pump timing lock pin is removed before continuing. Severe damage to the fuel injection pump will occur unless the fuel injection pump is unlocked.

Remove the fuel injection pump timing pin cap (2). Reverse the position of the timing pin (1) to unlock the pump and install the pin, cap and sealing washer.

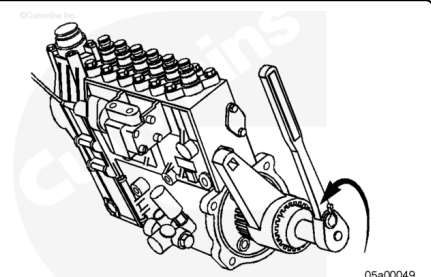
Torque Value: 30 n•m [22 ft-lb]



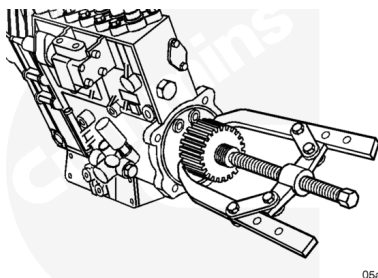
To remove the nut securing the spline adapter to the driveshaft, hold the spline adapter with service tool, Part Number 3163313, or equivalent.

Use a 41mm socket and long-handled ratchet or breaker bar to loosen the nut.

Remove the nut from the driveshaft.



To remove the driveshaft spline adapter from the tapered shaft of the fuel pump, use a standard two-jaw puller.

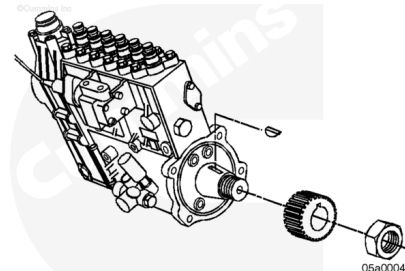


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Remove the spline adapter and Woodruff key.

Note : For an Open-Drive (Base-Mount) fuel pump, a woodruff key is **only** used on the rear of the shaft. When assembling the spline adapter to the pump, the woodruff key is **not** needed.

Note : The spline adapter **must** be removed and kept if the existing pump needs to be replaced or repaired. Replacement fuel injection pumps do **not** include spline adapters. **Only** an authorized Bosch™ repair location may service this fuel pump.



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Clean and Inspect for Reuse

⚠ WARNING ⚠

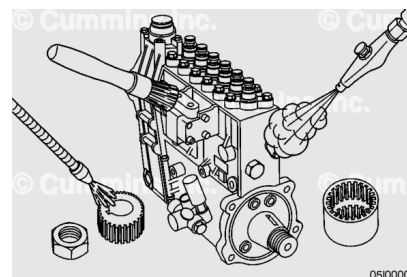
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Use a cleaning solution that won't hurt aluminum.

Use a brush and solvent around the areas where connections where the fuel pump were removed to make sure no dirt remains on the fuel injection pump.

Dry with compressed air.

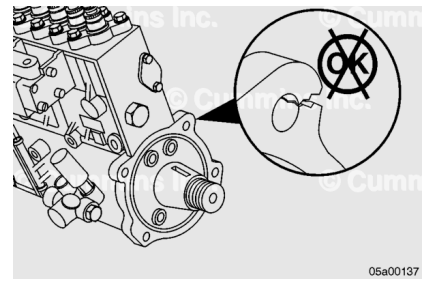


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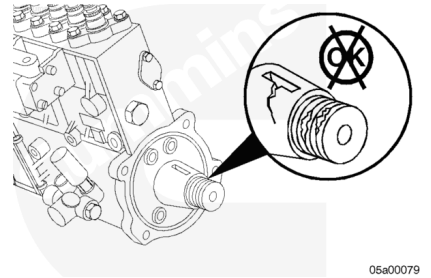
Inspect the fuel injection pump for the following damage:

- Mounting surfaces and capscrew holes for cracks and raised metal
- Damage to threads

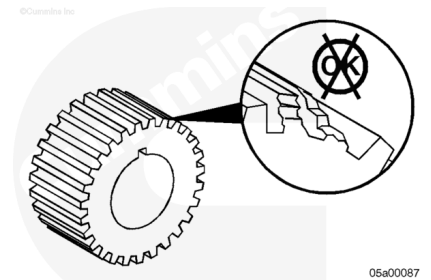
- Damaged cable or harness connectors.



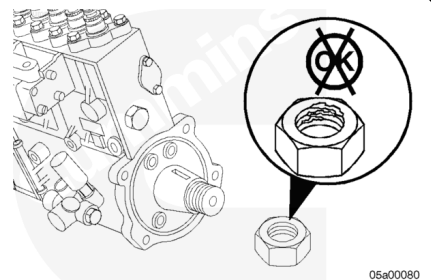
Inspect the fuel pump shaft and threads for damage.



Inspect the spline adapter for damage or missing splines.



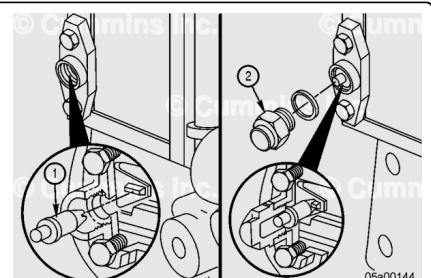
Inspect the nut threads and flats for damage.



Install

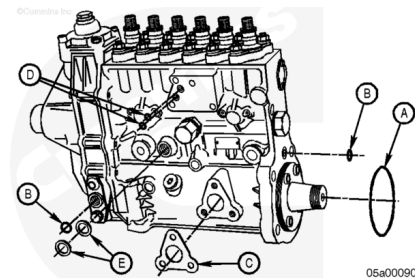
Remove the fuel injection pump timing pin cap (2). Verify the position of the timing pin. The fuel pump **must** be locked. Reverse the position of the timing pin (1) to lock the pump as shown.

Torque Value: 30 N·m [22 ft·lb]



Seal Replacement

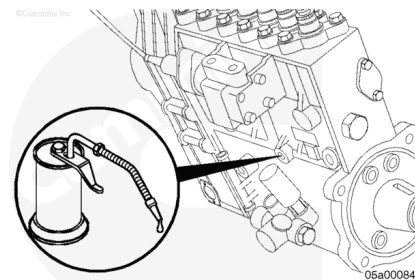
- A. O-ring seal (flange drive housing)
- B. O-ring seal (flange drive oil supply)
- C. Lift pump gasket
- D. O-ring seals - EHAB to injection pump
- E. Sealing washer (banjo connectors).



⚠ CAUTION ⚠

Failure to prelubricate the fuel injection pump can cause pump damage.

Prelubricate the fuel pump oil sump by adding clean engine oil to the fuel pump oil drain opening until oil comes out of the drain.



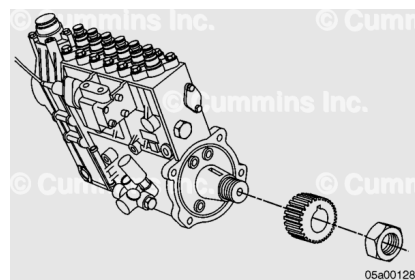
⚠ CAUTION ⚠

Check to make sure the injection pump timing lock pin is removed before continuing. Severe damage to the fuel injection pump will occur unless the fuel injection pump is unlocked.

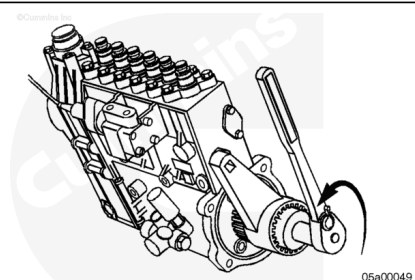
Prior to installation, the tapered fuel pump shaft and spline adaptor should be cleaned with non-petroleum based cleaner, (3824510) or equivalent, and wiped with a lint free rag. Contact with tapered shaft and inside of the spline adapter with bare hands should be avoided as it contaminates the mating surfaces.

Note : For an Open-Drive (Base-Mount) fuel pump, a Woodruff key is not required for installation of the spline adapter.

Slide the adapter onto the fuel pump shaft.



Install the nut and hand-tighten. Hold the spline adapter with service tool, Part Number 3163313 (with breaker bar attached), or equivalent, and tighten the nut, use a 41-mm socket.

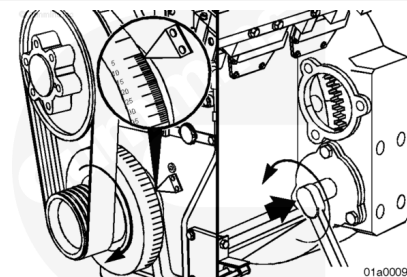


Pump Model	Torque Value	339 N•m	250 ft-lb
P8500			

Pump Model P7100	Torque Value	339 N•m	250 ft-lb
Pump Model RP39	Torque Value	475 N•m	350 ft-lb

⚠ CAUTION ⚠

Both the right- and left-bank fuel pumps must be removed/installed with the engine in this position. Severe engine damage can occur if static injection timing is not set properly.



⚠ CAUTION ⚠

Check that all packing caps are removed before installation.

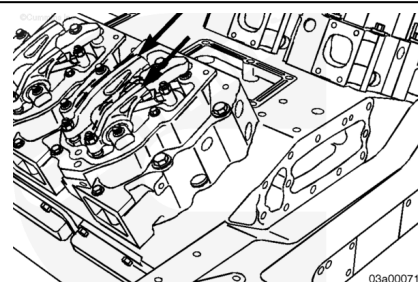
⚠ CAUTION ⚠

The pump needs to be locked at this point.

Rotate the engine slowly **clockwise** until the crankshaft damper pointer aligns with the specified engine timing, which is provided on the engine data plate, before Top Dead Center for the right-bank number 1 cylinder on the compression stroke.

Remove the right-bank number 1 cylinder rocker cover, and verify the number 1 cylinder is on the compression stroke.

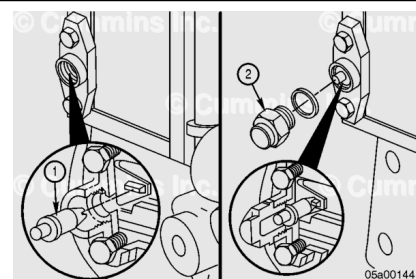
Note : Both of the rocker levers will be loose or the push rods will be the same height if the piston is on the compression stroke.



Verify the engine is in the correct position for fuel pumps to be installed on the engine.

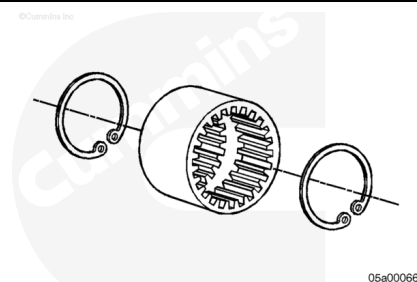
Remove the fuel injection pump timing pin cap (2). Verify the position of the timing pin. The fuel pump must be locked. Reverse the position of the timing pin (1) to lock the pump, as shown.

Install the timing pin, cap, and sealing washer.



Torque Value: 30 N•m [22 ft-lb]

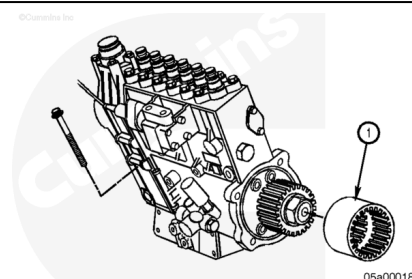
Install the retaining rings inside the spline coupling.



Install the spline coupling (1) onto the pump driveshaft gear.

Install the mounting bracket to the fuel pump. Make sure the dowel rings in the bracket align with the fuel pump.

Note : Different fuel pumps and mounting brackets are used between generator-drive and industrial engines.



Open-Drive (Base-Mount) Fuel Pump

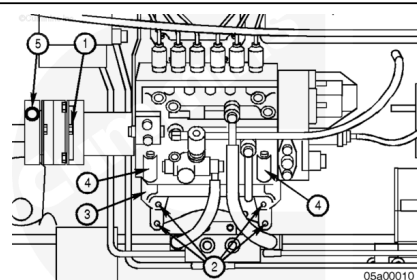
Tighten the four pump bracket mounting capscrews (4).

Set the fuel pump assembly in position.

With the bracket installed on the pump, set the fuel pump assembly at an angle, and install the assembly (3) on the engine. Make sure the dowel rings in the bracket align with the holes in the block.

Tighten the bracket capscrews (2). Tighten the coupling capscrew (1).

Note : For a Flange-Mount Fuel Pump do not tighten the capscrews.

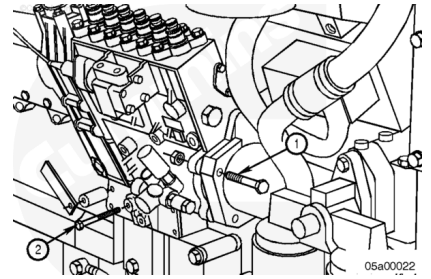


Bracket-Mounting Capscrews (2)	Torque Value	65 N•m	48 ft-lb
Coupling Capscrews (1)	Torque Value	108 N•m	80 ft-lb
Pump-Mounting Capscrews (4)	Torque Value	65 N•m	48 ft-lb
Pinch Bolt (5)	Torque Value	163 N•m	120 ft-lb

Note : Loctite™, Part Number 3824040 or equivalent, can be required to keep the shaft from slipping on the coupling. After installation, add a couple of drops to the pinch bolt slot.

⚠ CAUTION ⚠

If the pump does not align, do not force the pump and drive into alignment. Forcing will result in fuel pump damage. Use vegetable oil to lubricate the inside of the injection drive housing.



Flange-Mount Fuel Pump (P8500, P7100, and RP39)

Engage the spline coupling, and install the pump driveshaft gear into the drive housing. Install the flange-mounting capscrews and nuts (1) and the fuel-pump-mounting capscrews and washers (2).

Tighten the flange mounting capscrews (1).

Torque Value: 65 n·m [48 ft-lb]

Tighten the fuel pump mounting capscrews (2).

Torque Value: 65 n·m [48 ft-lb]

Note : The right-bank fuel pump bracket has six mounting bolts. The left-bank pump bracket has three mounting bolts. The flange-mount drive housing uses support brackets located off the number 2 camshaft follower covers for P8500 and P7100 fuel pumps. The Flange-Mount drive housing uses the ECM mounting bracket for RP39 fuel pumps.

⚠ CAUTION ⚠

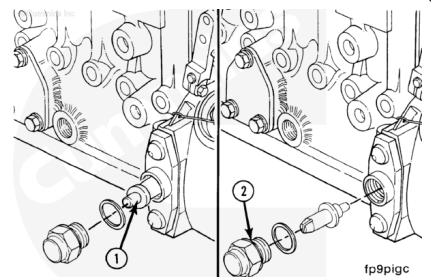
Severe damage to the fuel pump will occur unless the fuel pump is unlocked prior to operation.

Remove the fuel injection pump timing pin cap (2). Reverse the position of the timing pin (1) to unlock the pump and install the pin, cap, and sealing washer.

Set with the initial torque value.

Torque Value: 30 n·m [22 ft-lb]

Note : The final torque value should only be set if the pump is verified to still be in the correct timing position after the initial torque value is set.



Note : The timing pin is reversed when installing the on the engine.

Set the final torque value.

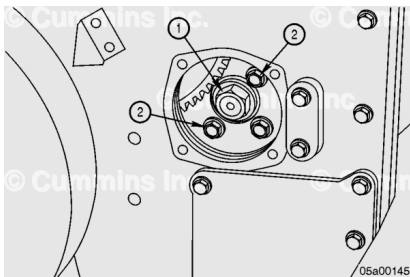
Torque Value: 30 n•m [22 ft-lb]

⚠ CAUTION ⚠

Use an evaporating cleaning solvent to clean the driveshaft tapered nose. Any lubricant present can allow timing to slip.

⚠ CAUTION ⚠

The timing pin must be removed before tightening the large nut (1) above 12 N•m [106 in-lb], or fuel pump damage will occur.



Install the drive gear and hub assembly on the shaft. Install the large nut (1) and washer securing the drive gear assembly to the drive shaft. Initial seating torque values for one-piece and two-piece assemblies:

Torque Value: 12 n•m [106 in-lb]

Verify the engine is still in time after seating the large nut (1).

Note : Check to make sure the drive gear assembly will be supported by the gear train. If not, use a wire to prevent it from dropping into the oil pan.

Note : A key is not installed on flange-mount fuel pump drive assemblies.

There are one-piece and two-piece drive gear assemblies. Tighten the nut (1) to the final torque value to secure the drive gear assembly to the driveshaft.

One-Piece Assembly	Torque Value	366 N•m	270 ft-lb
Two-Piece Assembly	Torque Value	434 N•m	320 ft-lb

Tighten the four capscrews (2) securing the drive gear to the hub on two-piece assemblies **only**.

Torque Value: 108 n•m [80 ft-lb]

Install the gear cover (if removed). Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-031.html)

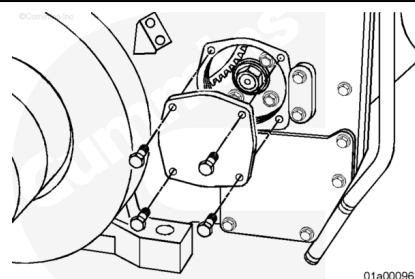
Note : One-piece drive gear assemblies have two capscrews, while the two-piece drive gear assemblies have four capscrews.

If equipped with a fuel pump drive gear cover plate, install the fuel pump drive gear cover plate o-ring.

Lubricate the o-ring with vegetable oil.

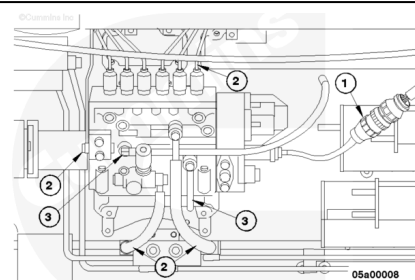
Tighten the capscrews.

Torque Value: 41 n•m [30 ft-lb]



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Install all the fuel pump oil lines (3).

Torque Value: 24 n•m [212 in-lb]

Install the high-pressure fuel lines (2). Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/57/57-006-024-tr.html)

Torque Value: 23 n•m [204 in-lb]

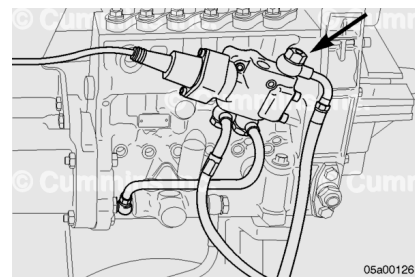
Install the low-pressure fuel lines (2). Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/57/57-006-024-tr.html)

Torque Value: 27 n•m [239 in-lb]

Install the electrical connection (1).

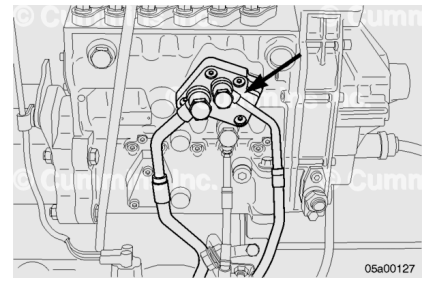
⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.



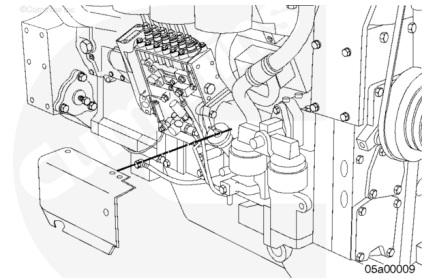
Install the overflow/pressure regulator valve onto the mechanical fuel pump. Refer to Procedure 005-066 in Section 5. (/qs3/pubsys2/xml/en/procedures/57/57-005-066.html)

Install the overflow/pressure regulator valve onto the electronic fuel pump. Refer to Procedure 005-066 in Section 5. (</qs3/pubsys2/xml/en/procedures/57/57-005-066.html>)



Install the open drive coupling cover.
Install the cover bracket and capscrews.

Torque Value: 60 n•m [44 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the electrical connection.
- Connect the batteries. Refer to the OEM service manual.
- Prime fuel system. Refer to Procedure 005-234 in Section 5. (</qs3/pubsys2/xml/en/procedures/57/57-005-234.html>)
- Start the engine and check for leaks.

